Projectile Motion I

Answer key

Conventions used throughout

• **Symbols:** v_1 = initial velocity, v_2 = final velocity, Δt = time interval, Δd = displacement. Subscript x means the horizontal part and y means the vertical part, so v_{1y} is the vertical part of the initial velocity. v_x has no 1 or 2 because it never changes.

• **Signs in the working:** +x is the way the object moves sideways (to the right in every diagram). +y is up, so gravity gives $a_y = -9.81$ m/s^2.

• **Directions in final answers:** [R] = right, [D] = down. [R 66° D] means: start pointing right, then turn 66° toward down.

• **Big idea:** split the motion into two separate problems. Sideways, no force acts, so v_x stays constant. Vertically, gravity acts, so $a_y = -9.81$ m/s^2. The only thing the two directions share is the time, Δt.

• g = 9.81 m/s^2, no air resistance. Every step keeps full precision; only final answers are rounded.

CONCEPT 1

Shape of the path

1. What is the shape of the path followed by the projectile?

Step 1 – Follow a ball launched sideways at 10 m/s

Sideways, nothing pushes the ball, so it moves the same 5.0 m every 0.5 s. Vertically, gravity makes it fall faster and faster. Here is the drop after 0.5 s:

$$\text{Formula: } \Delta d_y = v_{1y}\Delta t + \frac{1}{2}a_y \Delta t^2$$

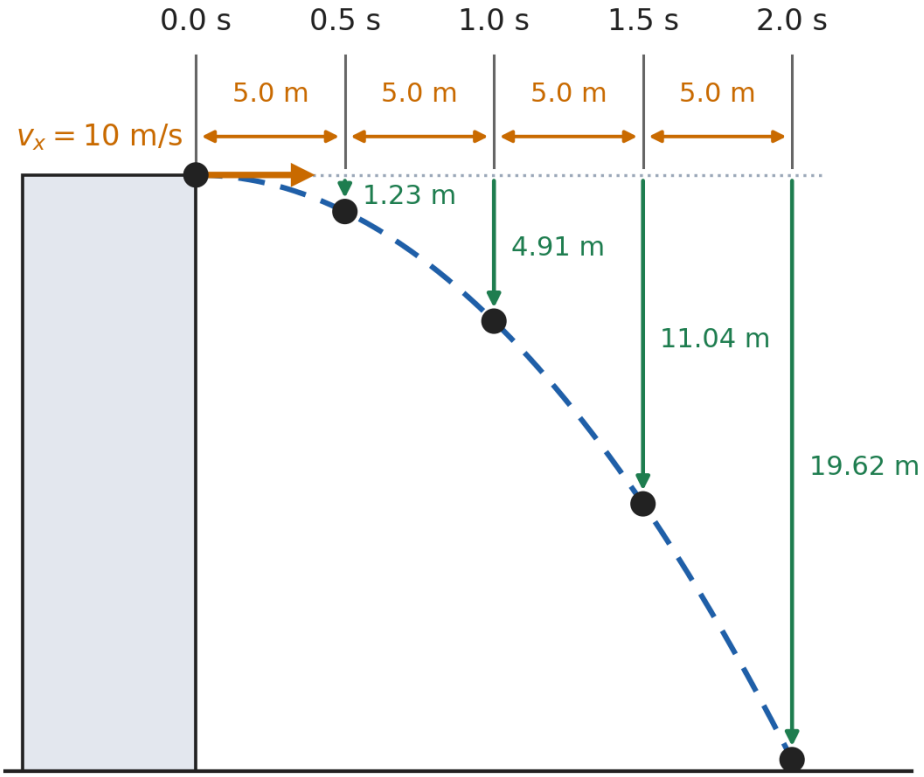
$$\text{Put in } v_{1y} = 0: \quad \Delta d_y = \frac{1}{2}a_y \Delta t^2$$

$$\text{Put in } a_y = -9.81,\ \Delta t = 0.5: \quad \Delta d_y = \frac{1}{2}(-9.81)(0.5)^2 = -1.23 \text{ m}$$

The minus sign means down, so the ball has dropped 1.23 m. The other times are done the same way:

Time Δt (s)	Sideways distance (m)	Distance dropped (m)
0	0	0
0.5	5.0	1.23
1.0	10.0	4.91
1.5	15.0	11.04
2.0	20.0	19.62

Diagram: where the ball is every 0.5 s (drawn to scale)



Step 2 – Read the pattern

Every 0.5 s the ball moves the same 5.0 m sideways, but it drops more each time (1.23 m, then 3.68 m more, then 6.13 m more, then 8.58 m more). Steady sideways motion plus faster and faster falling gives a curve that bends down more and more. That curve is a **parabola**.

FINAL ANSWER

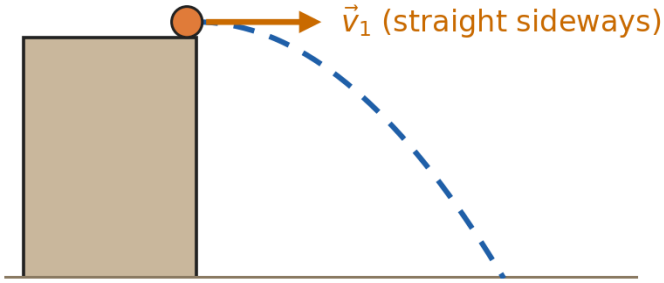
1. A parabola

CONCEPT 2

Horizontal launch

2. If a projectile is launched horizontally, what is its initial vertical velocity?

Picture the situation



“Launched horizontally” means the object leaves moving straight sideways.

Step 1 – Find the vertical part of the launch velocity

Straight sideways means the launch angle is 0°. The vertical part of a velocity uses sine:

Formula: $v_{1y} = v_1 \sin\theta$

Put in $\theta = 0°$: $v_{1y} = v_1 \sin 0° = v_1 \times 0 = 0$ m/s

It starts with no up or down motion. Gravity then builds up its downward velocity from zero.

FINAL ANSWER

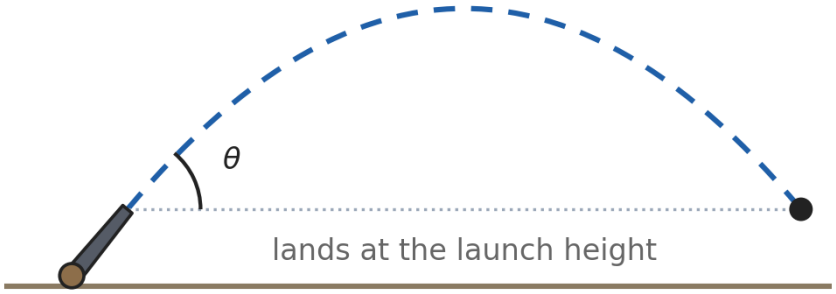
2. $v_{1y} = 0$ m/s

CONCEPT 3

Launch angle and distance

3. For a projectile that lands at the same level it's launched at, predict:
   a. What launch angle will give a projectile maximum horizontal distance?
   b. What other angle gives you the same distance as 30°? As 55°?
   c. What is the rule for finding two angles that give the same distance?

Picture the situation



Launched at an angle θ and landing at the same height it started from.

(a) What launch angle will give a projectile maximum horizontal distance?

Test it with numbers: launch at 10 m/s and find the distance for several angles. The full working for 45° is below. Every other angle uses the same three steps.

Step 1 – Split the launch velocity into parts

The sideways part is next to the angle, so it uses cosine. The vertical part is opposite the angle, so it uses sine.

Formula: $v_x = v_1 \cos\theta$

$v_x = 10\cos 45° = 7.07$ m/s

Formula: $v_{1y} = v_1 \sin\theta$

$v_{1y} = 10\sin 45° = 7.07$ m/s

Step 2 – Time in the air

At the top the ball stops rising for an instant, so $v_{2y} = 0$ there. Find the time to reach the top, Δt_{up}. The trip down takes just as long, because the ball lands at the height it started from, so the total time is double.

Formula: $v_{2y} = v_{1y} + a_y \Delta t$

Put in $v_{2y} = 0,\ v_{1y} = 7.07,\ a_y = -9.81$: $0 = 7.07 + (-9.81)\Delta t_{up}$

Add $9.81\,\Delta t_{up}$ to both sides: $9.81\,\Delta t_{up} = 7.07$

Divide both sides by 9.81: $\Delta t_{up} = \frac{7.07}{9.81} = 0.721$ s

Total time: $\Delta t = 2 \times 0.721 = 1.442$ s

Step 3 – Sideways distance

Sideways the speed stays at 7.07 m/s for the whole 1.442 s, so distance = speed × time.

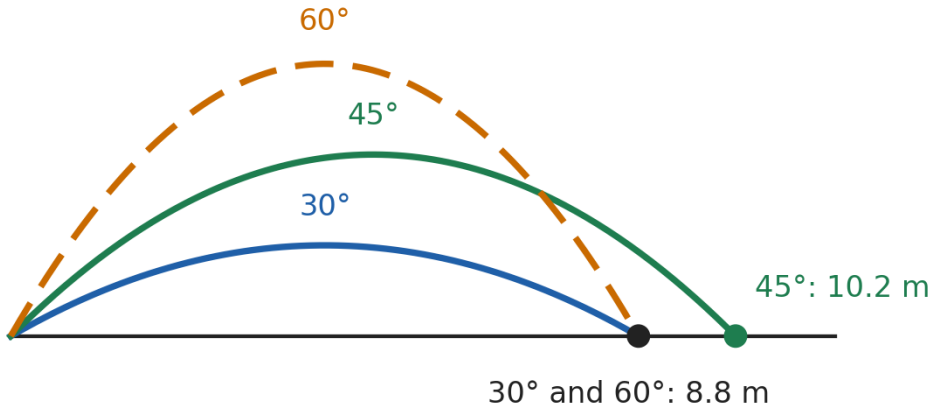
Formula: $\Delta d_x = v_x \Delta t$

$\Delta d_x = (7.07\text{ m/s})(1.442\text{ s}) = 10.19$ m

Results for every angle (10 m/s launch, same three steps):

Angle	v_x (m/s)	v_{1y} (m/s)	Time in air Δt (s)	Distance Δd_x (m)
15°	9.66	2.59	0.528	5.1
30°	8.66	5.00	1.019	8.8
35°	8.19	5.74	1.169	9.6
45°	**7.07**	**7.07**	**1.442**	**10.2 (largest)**
55°	5.74	8.19	1.670	9.6
60°	5.00	8.66	1.766	8.8
75°	2.59	9.66	1.969	5.1

Diagram: the 30°, 45° and 60° paths from the table (drawn to scale)



Step 4 – Pick the largest distance

The distance grows from 15° to 45°, then shrinks from 45° to 75°, so **45°** goes farthest. A low angle moves fast sideways but lands quickly; a high angle stays up longer but moves slowly sideways. 45° gives the best balance.

(b) What other angle gives you the same distance as 30°? As 55°?

30° and 60° both go 8.8 m. 35° and 55° both go 9.6 m.

(c) What is the rule for finding two angles that give the same distance?

30° + 60° = 90° and 35° + 55° = 90°. The two angles in a pair swap their speeds (30°: 8.66 m/s sideways, 5.00 m/s up; 60°: 5.00 m/s sideways, 8.66 m/s up). The higher shot is slower sideways but stays up longer, by exactly enough to land at the same spot:

$$30°: \quad \Delta d_x = (8.66\ \text{m/s})(1.019\ \text{s}) = 8.8\ \text{m}$$

$$60°: \quad \Delta d_x = (5.00\ \text{m/s})(1.766\ \text{s}) = 8.8\ \text{m}$$

FINAL ANSWER

(a) 45°

(b) 60° (same distance as 30°) and 35° (same distance as 55°)

(c) The two angles add up to 90°

CONCEPT 4

Horizontal and vertical motion

The diagram below shows the path of a projectile as at flies through the air

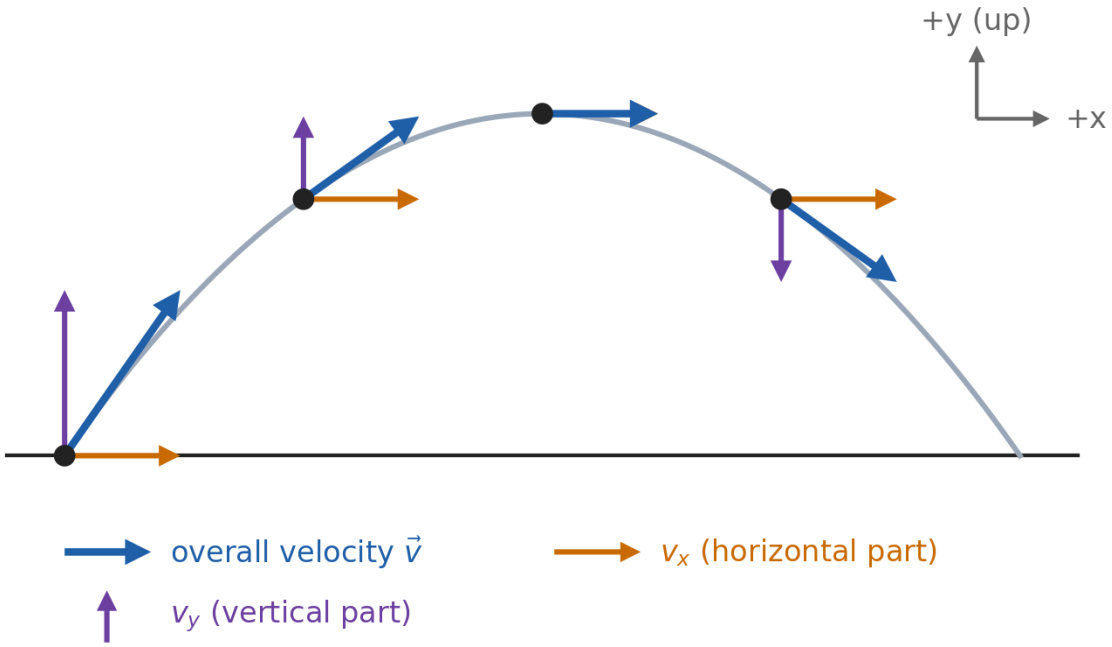
a. Describe the horizontal and vertical motion of a projectile.

b. Based on your answer to a, what is the only equation you will always use to analyze the horizontal motion of a projectile?

c. Focus on the overall velocity. (The central arrows are the overall velocity). Does it ever become zero? Why or why not?

d. Consider how the overall velocity is oriented to the path of the projectile. What word do we use to describe this orientation?

Diagram: the velocity and its two parts at four points (drawn to scale)



(a) Describe the horizontal and vertical motion of a projectile.

Horizontal: the orange arrows (v_x) are all the same length. No force acts sideways, so $a_x = 0$ and the horizontal motion is **uniform motion** (constant velocity).

Vertical: the purple arrows (v_y) shrink on the way up, are zero at the top, and grow downward on the way down. Gravity gives a constant $a_y = -9.81\ m/s^2$, so the vertical motion is **uniformly accelerated motion**.

(b) Based on your answer to a, what is the only equation you will always use to analyze the horizontal motion of a projectile?

The horizontal velocity never changes, so distance = velocity × time:

$$\Delta d_x = v_x \Delta t$$

(c) Focus on the overall velocity. (The central arrows are the overall velocity). Does it ever become zero? Why or why not?

The overall velocity is zero only if both parts are zero at the same time. At the top $v_y = 0$, but v_x never changes, so it is still there. The speed (the size of the velocity) comes from both parts:

$$\text{Formula: } |\vec{v}| = \sqrt{v_x^2 + v_y^2}$$

$$\text{At the top, } v_y = 0: \quad |\vec{v}| = \sqrt{v_x^2 + 0^2} = v_x$$

So the speed is smallest at the top, but not zero (unless the object is thrown straight up, so $v_x = 0$).

(d) Consider how the overall velocity is oriented to the path of the projectile. What word do we use to describe this orientation?

Each blue arrow points along the path in the direction of motion, just touching the curve without cutting across it. That is called **tangent**.

FINAL ANSWER

(a) Horizontal: uniform motion ($a_x = 0$)

Vertical: uniformly accelerated motion, $\vec{a} = 9.81\ m/s^2$ [D]

(b) $\Delta d_x = v_x \Delta t$

(c) No. v_x never changes, so the velocity is never zero

(d) Tangent: the velocity is always tangent to the path

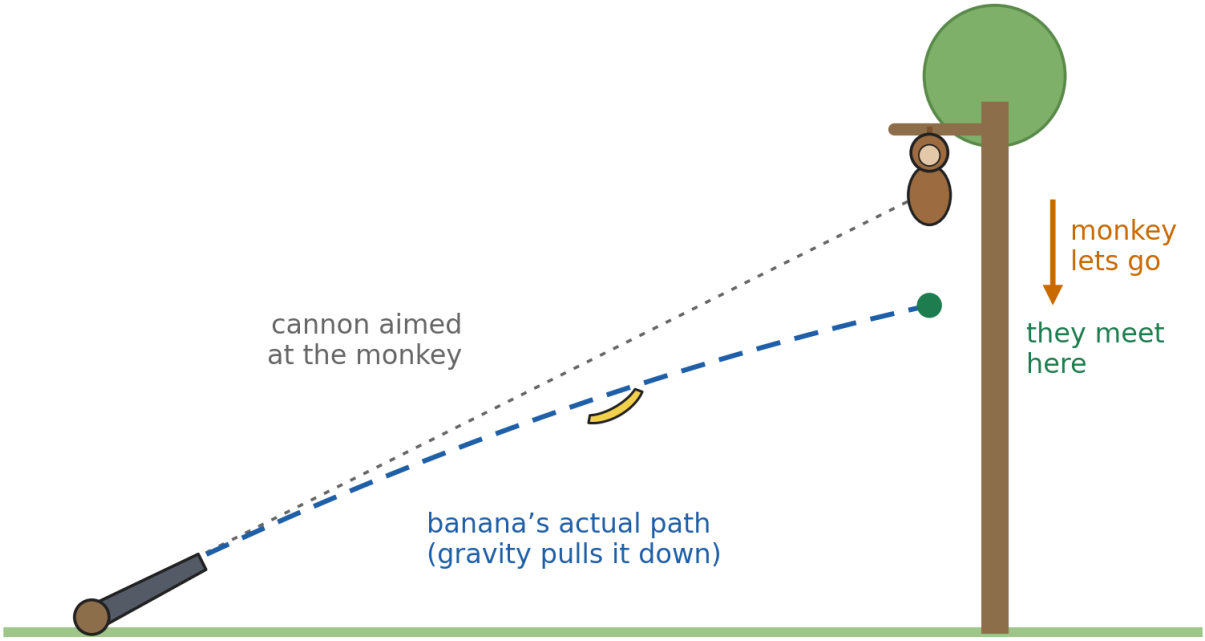
PROBLEM

Monkey and banana

In the shoot the monkey scenario, if the cannon fired the banana with the initial velocity of 30.0 m/s at an angle of 33.0° above the horizontal and the tree is 16.0 m tall and 25 m away from the cannon, how far does the monkey fall before it catches the banana? Answer this question by breaking it down into the steps below.

- Make a quick labelled sketch of the situation. Define positive x and y. Then fill in the table
- Why is it impossible to calculate anything with the vertical component right now?
- How can you use the x-component to help? Do it. Then go back and solve the vertical.

Picture the situation



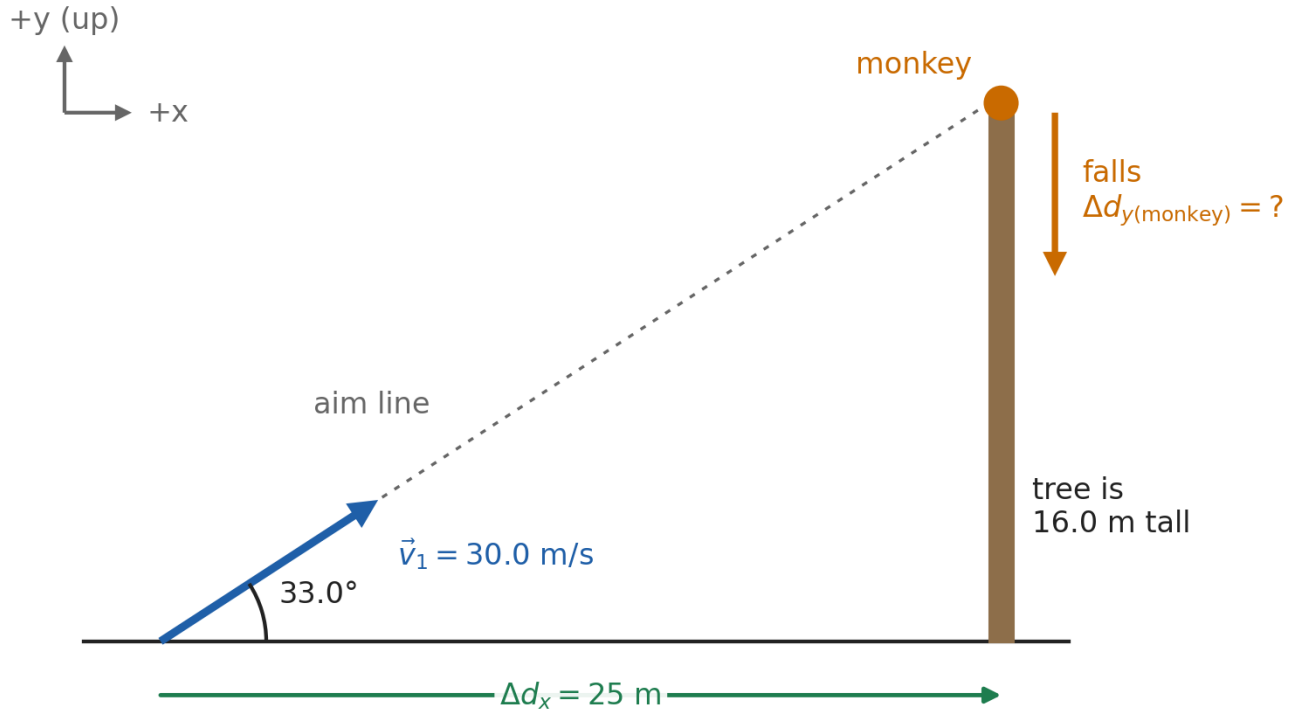
The cannon is aimed at the monkey. The monkey lets go the instant the cannon fires, and gravity pulls both the monkey and the banana down.

SOLVING FOR

How far the monkey falls, $\Delta d_{y(monkey)}$. Symbols marked (monkey) are for the monkey; all other symbols are for the banana.

Part 1 Make a quick labelled sketch of the situation. Define positive x and y. Then fill in the table

Labelled sketch: +x is toward the tree, +y is up (drawn to scale)



Step 1 – Split the launch velocity into parts

The table needs the horizontal and vertical parts of the launch velocity. The sideways part is next to the 33.0° angle (cosine); the vertical part is opposite it (sine).

Formula: $v_x = v_1 \cos\theta$

$v_x = 30.0 \cos 33.0° = 25.16$ m/s

Formula: $v_{1y} = v_1 \sin\theta$

$v_{1y} = 30.0 \sin 33.0° = 16.34$ m/s

Diagram: the launch velocity and its parts

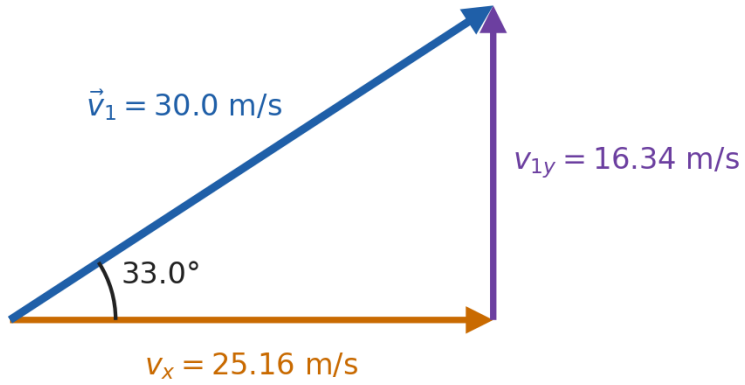


Table for the banana

	x (horizontal)	y (vertical)
Initial velocity	v_x = 25.16 m/s	v_{1y} = 16.34 m/s
Acceleration	a_x = 0	a_y = −9.81 m/s^2
Displacement	Δd_x = 25 m	Δd_y = ?
Time	Δt = ?	Δt = ? (the same Δt)

Part 2 Why is it impossible to calculate anything with the vertical component right now?

The vertical equation has two unknowns: Δd_y (the banana's height at the tree) and Δt (how long it takes to get there). One equation can't be solved for two unknowns.

$$\Delta d_y = v_{1y}\Delta t + \frac{1}{2}a_y\Delta t^2 \quad (\Delta d_y \text{ and } \Delta t \text{ unknown})$$

Part 3 How can you use the x-component to help? Do it. Then go back and solve the vertical.

Step 1 – Find the time from the horizontal motion

The x column is missing only Δt, so solve $\Delta d_x = v_x \Delta t$ for it. The banana reaches the tree at one moment, so this is also the time for the vertical motion.

Formula: $\Delta d_x = v_x \Delta t$

Divide both sides by v_x: $\Delta t = \frac{\Delta d_x}{v_x}$

$$\Delta t = \frac{25 \text{ m}}{25.16 \text{ m/s}} = 0.9936 \text{ s}$$

Step 2 – How far the monkey falls

The monkey lets go when the cannon fires, so it falls for the same 0.9936 s. It starts at rest, so $v_{1y(\text{monkey})} = 0$.

Formula: $\Delta d_y = v_{1y} \Delta t + \frac{1}{2} a_y \Delta t^2$

$$\Delta d_{y(\text{monkey})} = (0)(0.9936) + \frac{1}{2}(-9.81)(0.9936)^2 = -4.84 \text{ m}$$

The minus sign means down. The monkey ends up at 16.0 + (−4.84) = 11.16 m above the ground.

Step 3 – Check that the banana gets there

Use the same time in the banana's vertical equation:

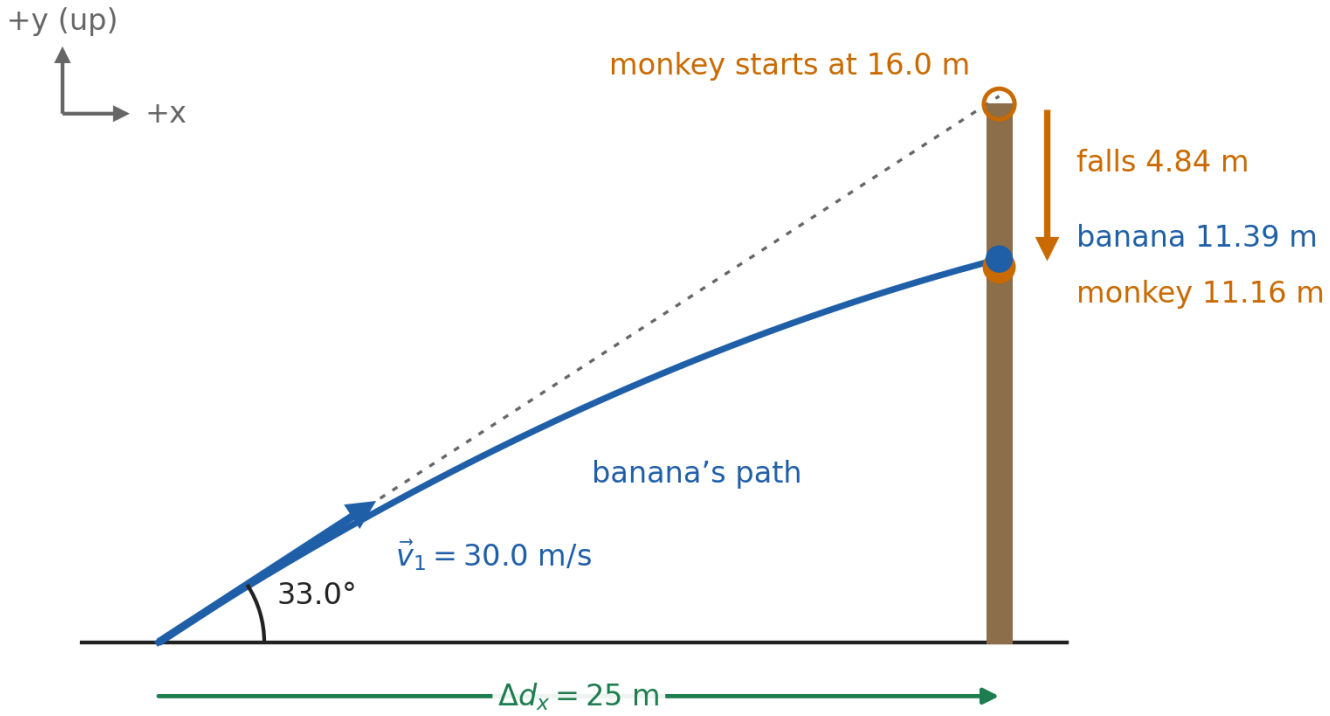
Formula: $\Delta d_y = v_{1y} \Delta t + \frac{1}{2} a_y \Delta t^2$

$$\Delta d_y = (16.34)(0.9936) + \frac{1}{2}(-9.81)(0.9936)^2$$

$$\Delta d_y = 16.235 + (-4.843) = 11.39 \text{ m}$$

Without gravity the banana would be at 16.235 m, on the aim line. Gravity pulls it 4.84 m below that, the same amount the monkey falls, so they meet. (The small 0.2 m gap is because 33.0° aims slightly above the monkey.)

Diagram: where the monkey and banana are when the banana reaches the tree



FINAL ANSWER

$\vec{\Delta d}_{monkey}$ = 4.8 m [D]

The monkey falls 4.8 m before it catches the banana.

EXAMPLE

Ball rolling off a counter

A ball is rolled at 2.2 m/s along a counter top that is 1.2 m above the floor. It rolls over the edge of the counter top.

a) How far horizontally does it travel before hitting the floor?

b) What is the velocity of the ball when it hits the floor?

SPLIT THE GIVENS INTO HORIZONTAL AND VERTICAL QUANTITIES. Time: is the same for both x and y.

Draw a fully labelled diagram showing the situation. All labels should indicate either x- or y-

Choose a direction to be the positive y-direction (up or down). Label it and make sure all your positives and negatives are correct.

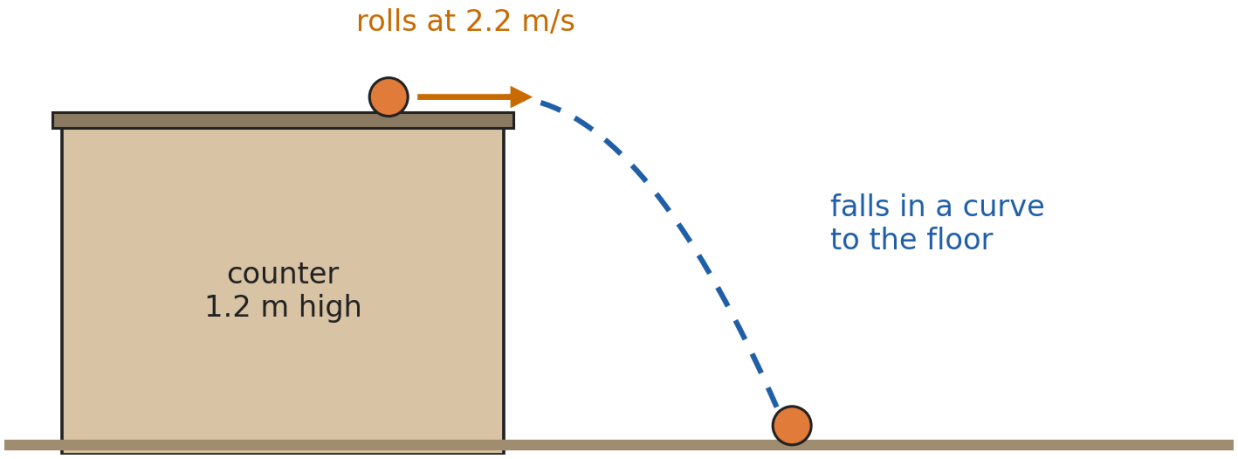
a) Horizontal motion is uniform. Write the equation for $\Delta\vec{d}_x$: _____

But you need to find the time first. Use the vertical motion:

b) The final velocity has two components. The horizontal velocity is _____. Now find the final vertical velocity:

Label the diagram and then find the overall final velocity:

Picture the situation

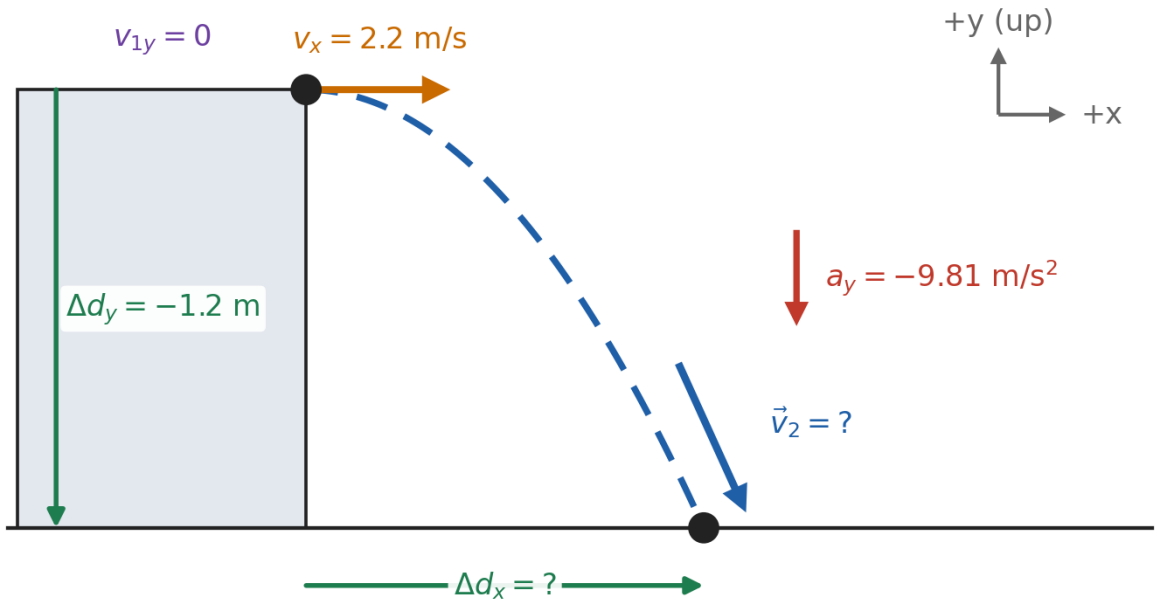


The ball leaves the edge moving straight sideways, then curves down to the floor.

Givens, split into x and y. +x is the way the ball rolls (right) and **+y is up**. The ball leaves the counter moving straight sideways, so $v_{1y} = 0$. The 1.2 m drop is down, so $\Delta d_y = -1.2$ m.

Givens	x (horizontal)	y (vertical)
Velocity	$v_x = 2.2$ m/s	$v_{1y} = 0$ m/s
Acceleration	$a_x = 0$	$a_y = -9.81$ m/s^2
Displacement	$\Delta d_x = ?$	$\Delta d_y = -1.2$ m
Time	$\Delta t = ?$	$\Delta t = ?$ (the same Δt)

Labelled diagram (drawn to scale)



(a) How far horizontally does it travel before hitting the floor?

Step 1 – The horizontal equation

Horizontal motion is uniform, so:

$$\Delta d_x = v_x \Delta t$$

This needs Δt, so find the time first from the vertical motion.

Step 2 – Find the fall time from the vertical motion

We know v_{1y}, a_y and Δd_y, so Δt is the only unknown. Time can't be negative, so take the positive square root.

Formula: $\Delta d_y = v_{1y}\Delta t + \frac{1}{2}a_y \Delta t^2$

Put in $v_{1y} = 0$: $\Delta d_y = \frac{1}{2}a_y \Delta t^2$

Multiply both sides by 2: $2\Delta d_y = a_y \Delta t^2$

Divide both sides by a_y: $\Delta t^2 = \frac{2\Delta d_y}{a_y}$

Square root both sides: $\Delta t = \sqrt{\frac{2\Delta d_y}{a_y}}$

$$\Delta t = \sqrt{\frac{2(-1.2\text{ m})}{-9.81\text{ m/s}^2}} = \sqrt{0.2446\text{ s}^2} = 0.4946\text{ s}$$

Step 3 – Sideways distance

Sideways, the ball keeps moving at 2.2 m/s for the whole 0.4946 s.

Formula: $\Delta d_x = v_x \Delta t$

$\Delta d_x = (2.2\text{ m/s})(0.4946\text{ s}) = 1.09\text{ m}$

(b) What is the velocity of the ball when it hits the floor?

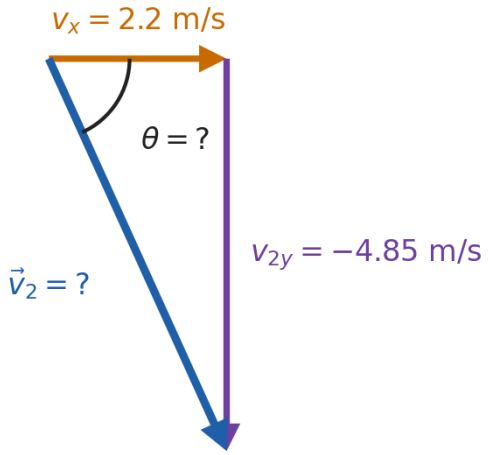
Step 1 – The two parts of the final velocity

The horizontal velocity never changes, so it is still v_x = 2.2 m/s. The vertical velocity starts at 0 and gravity changes it for 0.4946 s:

Formula: $v_{2y} = v_{1y} + a_y \Delta t$

$v_{2y} = 0 + (-9.81\text{ m/s}^2)(0.4946\text{ s}) = -4.852\text{ m/s}$

Diagram: the two parts of the final velocity



Step 2 – Combine the parts into one velocity

v_x and v_{2y} are at right angles, so $\vec{v}_2$ is the hypotenuse. Use the Pythagorean theorem for its size. For the angle θ below the horizontal, use tangent = opposite / adjacent with the sizes of the parts (no signs).

Formula: $|\vec{v}_2| = \sqrt{v_x^2 + v_{2y}^2}$

$$|\vec{v}_2| = \sqrt{(2.2)^2 + (-4.852)^2} = 5.33 \text{ m/s}$$

Formula: $\tan\theta = \dfrac{\text{opposite}}{\text{adjacent}} = \dfrac{|v_{2y}|}{v_x}$

$$\theta = \tan^{-1}\left(\frac{4.852}{2.2}\right) = 65.6°$$

θ is measured from the right, turning down, so the direction is [R 65.6° D].

FINAL ANSWER

(a) $\Delta\vec{d}_x$ = 1.1 m [R]

(b) $\vec{v}_2$ = 5.3 m/s [R 66° D]

Blank in (a): $\Delta\vec{d}_x = \vec{v}_x\Delta t$

Blank in (b): the horizontal velocity is 2.2 m/s [R] (unchanged)

Check: $v_{2y}^2 = v_{1y}^2 + 2a_y\Delta d_y = 0 + 2(-9.81)(-1.2) = 23.54$, so $|v_{2y}| = 4.85$ m/s, the same as Step 1.